Dataset Choice and Justification

This assignment uses the *Healthcare Stroke Prediction Dataset* obtained from Kaggle, which contains real-world patient health records such as age, gender, hypertension, heart disease, glucose level, BMI, smoking status, and employment type. The **target variable** indicates whether a patient has experienced a stroke. This dataset was selected because it represents a **non-trivial, real-world healthcare classification problem** and was not used in lectures or labs. Stroke prediction is a relevant and impactful application of supervised learning, where accurate classification can support early risk identification and preventive healthcare. The dataset's real-world nature and class imbalance make it suitable for evaluating different supervised learning algorithms beyond toy examples.

Model Implementation and Performance Evaluation

Three supervised learning models were trained and evaluated on the same dataset: a **Random Forest classifier** as a decision tree–based method, a **Support Vector Machine (SVM)**, and a **Multi-Layer Perceptron (MLP)** neural network. The data was preprocessed by handling missing values, encoding categorical variables, scaling numerical features, and splitting the dataset into training and test sets. Model performance was evaluated using **accuracy, F1-score, and confusion matrices**, ensuring a fair comparison across models. These metrics were chosen because the dataset is imbalanced, and accuracy alone may not fully reflect model effectiveness in detecting stroke cases.

Comparative Analysis of Results

The Random Forest model achieved the best overall performance, demonstrating higher accuracy and a more balanced confusion matrix compared to the other models. This can be attributed to its ensemble nature, which combines multiple decision trees to capture non-linear relationships and reduce overfitting. The SVM performed reasonably well but showed sensitivity to class imbalance, as it tends to favor the majority class in binary classification tasks. The MLP model was capable of learning complex patterns but required careful tuning and showed slightly less stable performance, likely due to the dataset size and imbalance. Overall, tree-based methods proved more robust for this healthcare dataset, while SVM and MLP performance depended more heavily on data distribution and preprocessing.

Ethical Considerations

In the context of healthcare, misclassification can have serious ethical implications. A **false negative**, where a patient at risk of stroke is predicted as healthy, could delay medical intervention and result in severe health consequences. Conversely, a **false positive** may cause unnecessary anxiety and lead to additional medical tests. Bias in the dataset, such as underrepresentation of certain demographic groups, may also result in unfair predictions. Therefore, machine learning models in healthcare should be used as **decision-support tools rather than replacements for medical professionals**, with careful consideration of fairness, transparency, and potential societal impact.